Taylor & Francis
Taylor & Francis Group

Research in Sports Medicine
An International Journal

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/gspm20

The management of a professional footballer recovering from a surgical repair of the proximal adductor longus tendon through a return-to-performance pathway. A case report

Andrew Mitchell & Craig Holding

To cite this article: Andrew Mitchell & Craig Holding (20 Sep 2024): The management of a professional footballer recovering from a surgical repair of the proximal adductor longus tendon through a return-to-performance pathway. A case report, Research in Sports Medicine, DOI: 10.1080/15438627.2024.2399005

To link to this article: https://doi.org/10.1080/15438627.2024.2399005

Published online: 20 Sep 2024.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=gspm20

RESEARCH IN SPORTS MEDICINE
https://doi.org/10.1080/15438627.2024.2399005

Routledge
Taylor & Francis Group

Check for updates

The management of a professional footballer recovering from a surgical repair of the proximal adductor longus tendon through a return-to-performance pathway. A case report

Andrew Mitchell[a] and Craig Holding[b]

[a]Medical Department, RasenBallsport Leipzig GmbH, Leipzig, Germany; [b]Irish Football Association, Belfast, Northern, Ireland

ABSTRACT
This case report details the management of a 23-year-old professional footballer who sustained a rupture of the proximal adductor longus (AL) tendon. Following surgical reattachment of the tendon, the player completed an eleven-phase return to performance (RTP_{erf}) pathway designed to ensure a rapid and safe return to play (RTP_{lay}). The pathway uses distinct phases that incorporate clinical, performance, and sport-specific criteria to guide decision-making throughout the process. This case report outlines the phases and criteria used in conjunction with shared decision-making by the interdisciplinary team (IDT) to ensure a successful RTP_{lay}. The effectiveness of this pathway was demonstrated by the player's return to competitive play 12 weeks post-surgery.

ARTICLE HISTORY
Received 24 June 2024
Accepted 27 August 2024

KEYWORDS
Criteria based; return to performance pathway; professional football; surgical repair; adductor longus tendon

Introduction

Groin injuries are extremely common in football, accounting for 18% of all injuries (Mosler et al., 2018). In addition, the AL muscle is reported to be involved in two-thirds (Mosler et al., 2018) of all cases and 17% of these are classified as grade 3 injuries (rupture or avulsion) (Serner et al., 2018). For grade 3 injuries, both conservative and surgical management have provided solid RTP_{lay} outcomes. Conservative management potentially provides shorter RTP_{lay} times (10–12 weeks) (Dimitrakopoulou et al., 2008; Ueblacker et al., 2016) and is better suited for small tendon retraction lengths (<2 cm), but carries the risk of uncontrolled tendon healing, weakness and residual pain (Best et al., 2020). In contrast, surgical management results in longer RTP_{lay} times (12–16 weeks) (Best et al., 2020), but is better suited to high-performance athletes that have suffered larger tendon retraction lengths (>2 cm) and/or damage to other structures such as the pyramidalis muscle, the anterior pubic ligament, and in some cases the pectineus muscle (Lempainen et al., 2021).

The purpose of this case report is to describe how a professional footballer successfully progressed through an RTP_{erf} pathway following a proximal AL tendon reattachment. This report will use the previously cited RTP_{erf} pathway, which has 11 clear phases (Figure 1) with clear progressions and objective criteria to help inform a shared decision-making

CONTACT Andrew Mitchell andrew.mitchell@redbulls.com Medical Department, RasenBallsport Leipzig GmbH, Cottaweg 3, Leipzig 04177, Germany

© 2024 Informa UK Limited, trading as Taylor & Francis Group

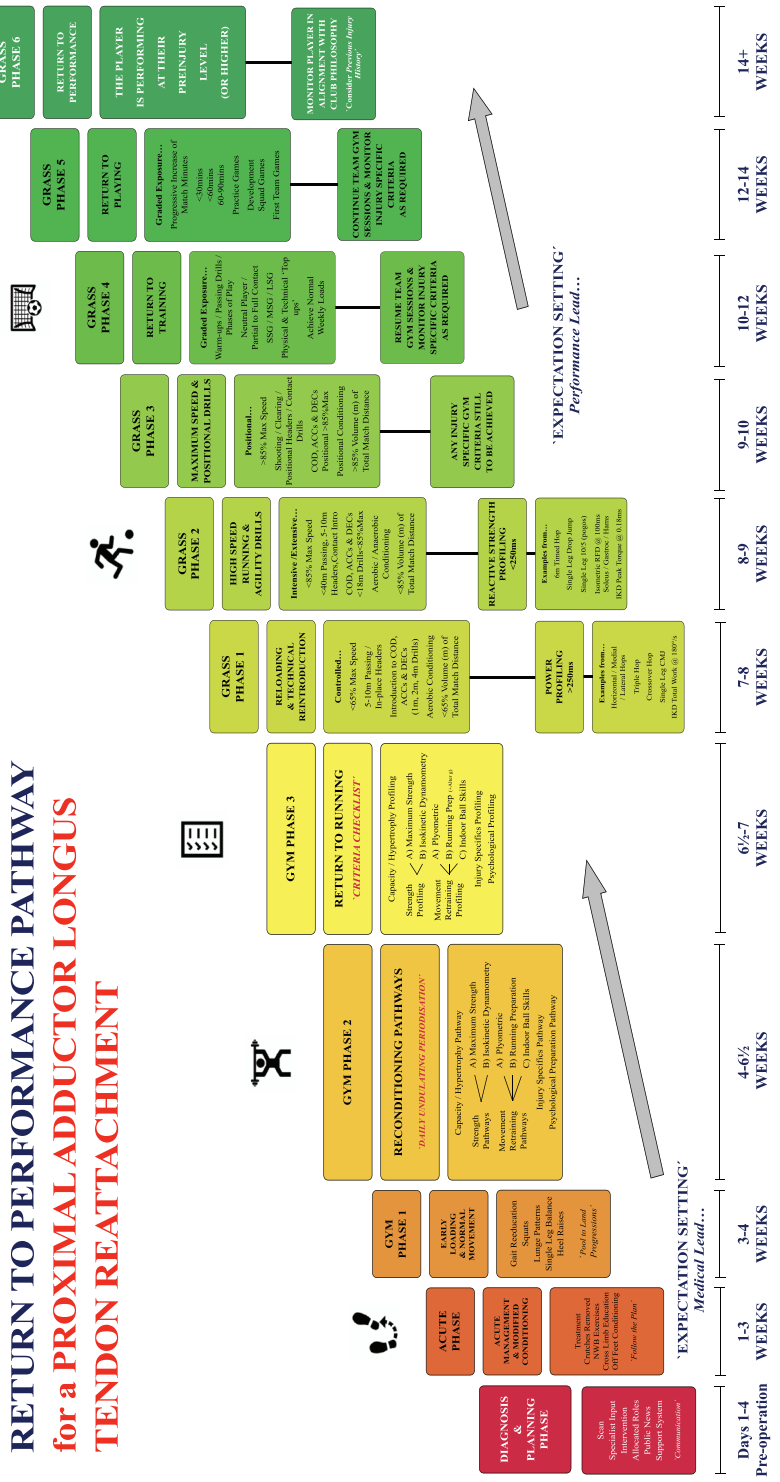

Figure 1. Shows the return-to-performance pathway for a proximal adductor longus tendon reattachment. Abbreviations: ACCs, accelerations; CMJ, counter-movement jump; COD, change of direction; DECs, decelerations; Hams, hamstrings; LSG, large-sided games; M, metres; Max, maximum; MSG, medium-sided games; Mins, minutes; Ms, milliseconds; NWB, non-weight bearing; RFD, rate of force development; SL, single leg; SSG, small-sided games; non-weight bearing; RFD, rate of force development; SL, single leg; SSG, small-sided games.

process (Mitchell & Gimpel, 2024). The pathway has three distinct gym-based phases to prepare the player for the subsequent grass phases. In conjunction with the grass phases, the player will also complete more advanced gym profiling (power and reactive strength) to ensure he is ready to resume team training (Grass Phase 4) and ultimately RTP_{lay} (Grass phase 5). The RTP_{erf} pathway is completed (Grass Phase 6) when the player is deemed to be performing at or above his previous level of performance.

Discussion

Case presentation

This previously healthy 23-year-old professional footballer was playing centre-back in a game when he attempted to clear a ball with his left foot. He immediately experienced an audible "pop" and pain in the adductor region of the kicking limb. The injury mechanism was established as abduction, extension and external rotation of the hip prior to kicking the ball. This particular movement places a high eccentric load on the AL (Charnock et al., 2009), with this particular mechanism accounting for 29% of all AL ruptures (Serner et al., 2018). After attempting the clearance, the player was unable to continue and was immediately removed from the field for clinical evaluation.

From the outset, the player presented with common signs and symptoms of a proximal AL tendon rupture. Initially, clinical assessment revealed a palpable defect/dip, swelling, and reproduction of pain in the region of the adductor origin on outer range resistance testing (Abate et al., 2023). Within 72 hours, the player underwent Magnetic resonance imaging (MRI), which revealed a full-thickness avulsion of the AL origin and fibrocartilage footprint. The tendon and fibrocartilage retracted distally 3.9 cm. Accompanying this report, the player had developed an antalgic gait and gross swelling in the adductor region. Compared to pre-injury metrics, he was significantly impaired in hip range of movement (flexion, abduction and internal rotation), and strength output (resisted adduction) at 0°, 45° and 90° of hip flexion. A shared decision-making process was undertaken by key stakeholders (doctors, orthopaedic surgeon, physiotherapists, and coaches), with the player (Dijkstra et al., 2017) electing for a proximal AL tendon reattachment with a RTP_{lay} goal of 12–16 weeks (Best et al., 2020).

Surgical management and acute phase

Five days post-injury the player underwent surgery, where a 3–5 cm incision was made 3 cm distal to the pubic tubercle over the adductor region. The retracted AL tendon was located and reattached to the pubic tubercle with 2 suture anchors. The subcutaneous tissue was then closed with absorbable sutures and the skin was closed with non-absorbable sutures (Lempainen et al., 2021). Immediately after surgery, appropriate nutritional support, wound care, rest, and the use of cryotherapy were priorities. At 7 days postoperatively, the player was able to stop using crutches as he was able to adopt a normal gait pattern and started a modified conditioning programme in the gym for the upper body and the contralateral limb.

Gym phase 1 - early loading and normal movement

During weeks 3–4 the player completed Gym Phase 1, regaining confidence in normal movements such as squats and basic lunge patterns, first in the pool and then on land. In the gym, regaining hip range of motion was prioritized along with the introduction of adductor exercises using ball squeezes at 0°, 45° and 80–90° of hip flexion (Thorborg, 2023). These isometric exercises are believed to be beneficial for tendon loading and pain reduction while promoting function (Hölmich et al., 1999). Prophylactic core stability exercises were also introduced including basic plank positions, standing squat positions and single leg (SL) balance-type exercises. Throughout this and subsequent phases, the player used a numeric pain rating scale (0–10) to rate their pain, with 0 being no pain and 10 being the worst perceived pain. A score of 2 or less was considered acceptable for exercise progression (Thorborg, 2023). Complementary to this, a global confidence rating was used by the player to guide progression and ensure ongoing communication between the player and the IDT (0% being no confidence and 100% being complete confidence, with 80% required for progression) (Mitchell & Gimpel, 2024).

Gym phase 2 – reconditioning pathways

Once the player had demonstrated the basic movement patterns of Gym Phase 1, they began Gym Phase 2, which focused on reconditioning pathways (Figure 1). This phase lasted two and a half weeks as the player progressed through the various reconditioning pathways in preparation for the return to running profiling in Gym Phase 3. Crucially, adductor rehabilitation was a priority because of the surgical reattachment and its role in football-related movements such as kicking and change of direction activities. Adductor exercises included concentric and eccentric banded adduction, abduction, hip flexion, and an introduction of multiplane combinations (Serner et al., 2020). Similarly, short lever Copenhagen and standing slide board exercises were performed as pain and confidence allowed (Serner et al., 2014). These, and the player's overall reconditioning, were supported by exposure to the other pathways of this phase (Figure 1).

The running preparation pathway included exercises such as wall drives, marching patterns and low-level sled pushes. Player engagement and confidence was enhanced by the indoor ball skills pathway, which used a stepwise approach to grading ball exercises from seated foot contacts to 1-2 m passes, in this case. The plyometric pathway developed the players' ability to absorb and produce force through the stretch-shortening cycle (SSC). This started in the pool with first double leg, and then SL exercises (squat jumps, to countermovement to drop jump techniques), and progressed on a soft surface and, towards the end of the phase, the normal gym floor (Mitchell et al., 2023).

As with all the exercises in this phase, linear actions were performed first, followed by the introduction of multidirectional patterns to gradually load the adductor region. The injury specifics pathway inevitably included adductor progressions but also the development of lumbo-pelvic control. Emphasis was placed on gluteal activation, avoidance of anterior pelvic tilt and control of trunk stability to optimize load distribution through the hip and groin region (King et al., 2018). Having progressed along the reconditioning pathways of Gym Phase 2, the player was profiled in the return to running criteria (Gym Phase 3) at weeks 6.5–7 to determine his readiness to run.

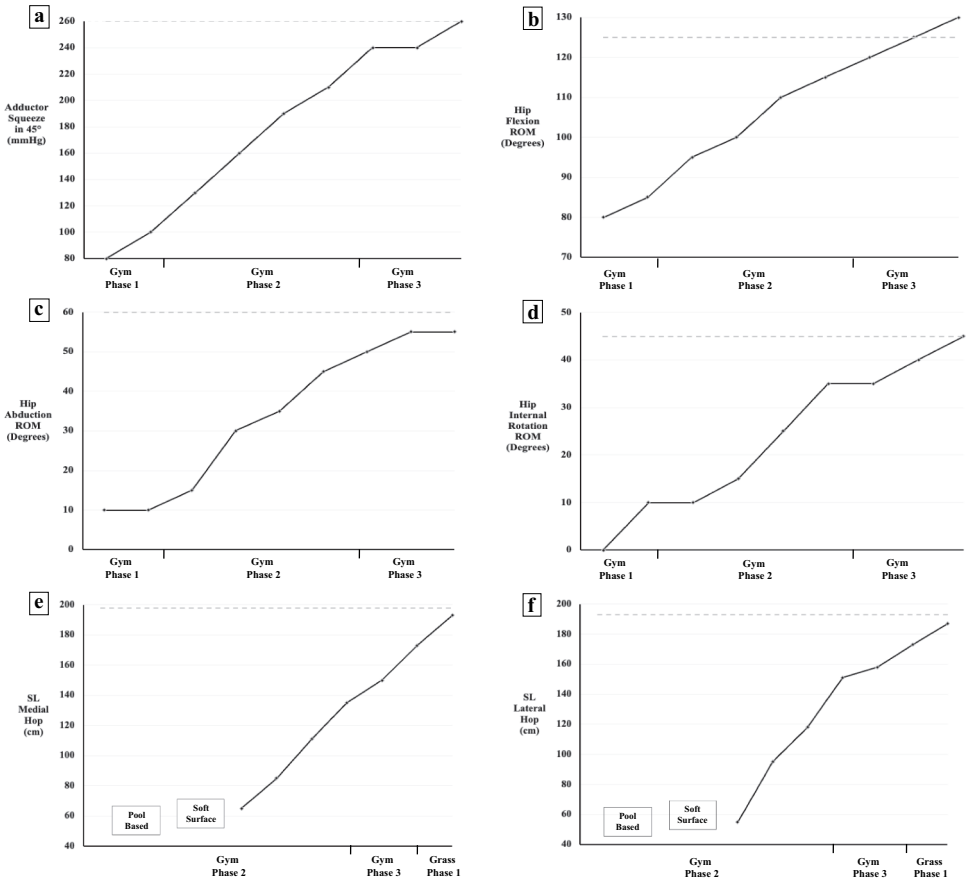

Figure 2. Demonstrates the monitoring of key objective criteria during the return-to-performance pathway. a, adductor squeeze (mmHg) b, hip flexion ROM (degrees); c, hip abduction ROM (degrees) d, hip internal rotation ROM (degrees). e, SL medial hop (cm). f SL lateral hop (cm). Black line recovering limb, grey line preinjury baseline measurements. Abbreviations: cm, centimetres; SL, single leg.

Gym phase 3 – return to running `criteria checklist´

From the RTP_{erf} pathway (Figure 1), Gym Phase 3 specifically profiles the player's ability to begin running and the introduction of sport-specific actions in Grass Phase 1. Minimum standards surrounding the theme of return to running have been previously documented (Mitchell & Gimpel, 2024) and were sufficiently met by this player during weeks 6.5–7. For example, the player completed low-intensity running preparation sessions consisting of A-skips, B-skips, butt kicks, and ankle bounds etc., in addition to an antigravity treadmill session at 95% of the player's body weight. The SSC was further evaluated by plyometric profiling, which included a hop battery (profiled by distance achieved) and force plate jumps (profiled by eccentric, concentric and performance variables such as jump height and reactive strength index) (Mitchell et al., 2021, 2022). Player confidence, movement proficiency and achieving 80% of pre-injury metrics were sufficient as 100% recovery of

SSC qualities is not possible at this point in the process. Injury specific criteria such as adductor squeeze and hip range of motion were all within 10% of pre-injury metrics (Figure 2), which was further supported by players' pain levels not exceeding 2/10. At 7 weeks post-surgery, an IDT decision based on the available date allowed the player to progress to Grass Phase 1.

Grass phase 1 – reloading and technical reintroduction

At 7 weeks, the player began Grass Phase 1 (reloading and technical reintroduction), which included the reintroduction of running, low-intensity sport-specific skills and the completion of power profiling in the gym (Figure 1). Using live data monitoring (Global Positioning System [GPS]), speeds were limited to <65% of a player's maximum and included a combination of pitch length runs (100 m), box-to-box runs (70 m) and football specific dribbling circuits such as the Hoff Drill (Mitchell et al., 2023). Similarly, accelerations (ACCs) and decelerations (DECs) were introduced (<65% of his maximal intensity) but were confined to small spaces (≤4 m) to limit any exposure to high-speed running (HSR) or high-intensity ACCs and DECs (Harper et al., 2022). In the gym, the player continued to develop his power qualities and was profiled through the hop battery (SL horizontal and triple hop techniques) and force plate (SL countermovement and drop) jumps. Specific to this injury, underlying strength (adductor/abductor ratio and outer range adductor strength, obtained from isometric profiling) (Mosler et al., 2017; Serner et al., 2020), and dynamic power (SL medial and lateral hop techniques) were targeted in conjunction with kicking actions using a resistance band. The normalization of these, together with a meticulous progression of kicking loads (2 m to 10 m distances) on the grass, improved player confidence, motivation and skill retraining.

Grass phase 2 – high speed running and agility drills

At 8 weeks, the player progressed to Grass Phase 2, which involved completing HSR and intensive agility drills, including exercises with reactive decision-making. Running speeds were increased to <85% of the player's maximum speed, and the intensity of ACCs and DECs was further increased by expanding the drill distances (Figure 1) (Harper et al., 2022). The kicking distance was systematically increased from 10 m to 40 m while closely monitoring the clinical signs, symptoms, and players' ratings of self-confidence. In the gym, reactive strength was profiled using the SL drop jump reactive strength index. Additionally, the adductor strength was specifically targeted for recovery from this injury. The minimum standards that needed to be met before entering Grass Phase 3 were an inner range adductor/abductor strength ratio of 1.2 (±0.2) N (Mosler et al., 2017), and an outer range isometric adductor strength >3.0 (±0.7) N·m/kg (Serner et al., 2020) in addition to other sports-specific grass and gym-based metrics (Table 1) (Mitchell & Gimpel, 2024).

Grass phase 3 – maximum speed and positional drills

From weeks 9–10, the player completed the maximum speed and positional drills of Grass Phase 3. Drills were often enhanced by the support of a technical coach or additional players to replicate exercises that were chaotic and highly variable by design. Defending drills

Table 1. The grass and gym-based metrics collected during gym phase 3 (return to running) and the subsequent grass phases.

	PHASE		GYM PHASE 3	GRASS PHASE 1	GRASS PHASE 2	GRASS PHASE 3	GRASS PHASE 4	GRASS PHASE 5	GRASS PHASE 6
			Return To Running Criteria Checklist	Reloading & Technical Reintroduction	High Speed Running & Agility Drills	Maximum Speed & Positional Drills	Return To Training	Return To Playing	Return To Performance
	METRICS		6-7 WEEKS	7-8 WEEKS	8-9 WEEKS	9-10 WEEKS	10-12 WEEKS	12-14 WEEKS	14 WEEKS +
GRASS CRITERIA	Total Metres (in one session)		2500 (Antigravity Treadmill)	5470 (48)	8188 (72)	10345 (92)	9927 (88)	9735 (86)	10316 (91)
	HSR Meters (>19,8 Km/h)		0 (0)	18 (3)	361 (55)	599 (92)	786 (121)	577 (89)	551 (85)
	Peak Speed (Km/h)		12,Km/h (35)	21 (60)	27,8 (80)	32,8 (94)	34 (97)	33,5 (96)	34,8 (100)
	Sprint meters (>25,2 Km/h)		0 (0)	0 (0)	39 (45)	89 (103)	249 (290)	84 (98)	70 (81)
	Accelerations Medium		Player Completed Indoor Balls Skills & ABC's Running Preparation Sessions	8 (9)	63 (68)	88 (96)	94 (102)	76 (83)	73 (79)
	Peak Acceleration (m/s²)			3,2 (62)	4,1 (79)	5,0 (96)	5,1 (98)	5,2 (100)	5,1 (79)
	Decelerations Medium			12 (11)	52 (48)	118 (118)	115 (106)	125 (115)	115 (106)
	Peak Deceleration (-m/s²)			-3,9 (57)	-5,1 (75)	-6,4 (94)	-6,7 (99)	-6,6 (97)	-6,5 (96)
	No of Sessions in a Week		4	4	5	5	5	5	5
GYM CRITERIA	Horizontal Hop (cm)	L	161 (74)	201 (93)	214 (99)	212 (98)	213 (98)	212 (98)	216 (100)
		R	212 (98)	214 (99)	215 (99)	209 (98)	214 (99)	217 (100)	215 (99)
	Triple Hop (cm)	L	550 (79)	604 (86)	648 (93)	689 (98)	701 (100)	695 (99)	698 (100)
		R	685 (98)	680 (97)	690 (99)	702 (98)	689 (98)	701 (102)	697 (101)
	SL CMJ (jump height - cm)	L	23 (77)	25 (83)	28 (93)	28 (93)	29 (97)	**28** (97)	30 (103)
		R	30 (100)	29 (97)	28 (93)	30 (83)	31 (103)	29 (97)	30 (100)
	SL Drop Jump (jump height - cm)	L	19 (70)	21 (78)	23 (85)	25 (93)	26 (96)	25 (96)	26 (100)
		R	26 (96)	27 (100)	28 (104)	26 (96)	27 (100)	26 (96)	27 (100)
	SL Drop Jump RSI (m/s)	L	0,45 (57)	0,56 (71)	0,68 (86)	0,68 (86)	0,70 (89)	0,71 (100)	0,70 (99)
		R	0,77 (97)	0,81 (103)	0,78 (99)	0,75 (95)	0,77 (97)	0,75 (95)	0,78 (99)
	Ab / Ad Strength Ratio (N)	L-R	0,91	1,0	1,11	1,2	1,2	1,2	1,2
	Outer Range Adductor Strength (N·m/kg)	L	2,9	3,3	3,9	3,9	4	4,1	4,1
		R	3,5	3,7	3,9	4,1	4,2	4,2	4,2

Scores in brackets for the grass criteria are the absolute values attained, expressed as a percentage of match values. Scores in brackets for the gym-based criteria are the left leg (injured leg) scores expressed as a percentage of the highest preinjury score of either the right or left leg. Abbreviations: cm, centimetres; N, newtons; N.m/kg, newtons per kilogram of body mass; m/s, metres; RSI, reactive strength index.

involved long passing, clearing actions and blocking movements to enhance player confidence in the injury. This was matched physically by the player achieving 94% of peak speed, 96% of peak ACC intensity, and 84% of peak DEC intensity (Table 1). In the gym, he continued to work on hip and pelvic strength exercises (King et al., 2018) and adductor-specific exercises (Serner et al., 2020), (Table 1) seeking further improvements and for prophylaxis. At the end of Grass Phase 3, in addition to physical characteristics, the player expressed a confidence level of over 90% in resuming team training (Grass Phase 4) in week 10.

Comparative outcomes

At 10 weeks, the IDT decision permitted the player to return to team training (Grass Phase 4) and was reintroduced to passing drills, phases of play and small (4 vs 4), medium (5 vs 5 to 8 vs 8) and large practice games (9 vs 9 to 11 vs 11). The player was also exposed to peak running speeds (97%) and peak ACCs and DECs intensities (98% and 99%, respectively) in training situations, helping him to replicate match demands. Subsequently, at 12 weeks, he completed 45 min in a U23 match without any complications (Grass Phase 5). Further match exposure was increased over the next 2 weeks, allowing the player to participate fully in the regular match schedule from 14 weeks (Grass Phase 6). Complementing these grass progressions was the continuity of gym-based strength, power and reactive strength exercises, helping to mitigate the risk of future injury (Lauersen et al., 2014).

Conclusion/Clinical implication

This case report demonstrates a successful RTP_{erf} for a professional footballer who sustained a rupture of the proximal AL tendon. Following an acute injury, the player was effectively managed through surgical reattachment and adherence to a progressive rehabilitation pathway. During the process, the player followed the RTP_{erf} pathway, which required him to meet phase-based clinical, psychological, and sports-specific criteria. The RTP_{erf} pathway was underpinned by the IDT's shared decision-making, which aimed to ensure a positive outcome and successful RTP_{erf}.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Patient/Data sharing involvement

Patient Signed Consent.

Author contributions

All authors take full responsibility for the writing, drafting and revisions of the manuscript. No other external contributors are involved.

References

Abate, M., Sammarchi, L., Calà, R., Milesi, G., Poerio, C. S., Del Vescovo, R., Corvino, A., Delli Pizzi, A., Cocco, G., & Salini, V. (2023). Isolated adductor longus avulsion in a young semi-professional football player: Imaging contribution and therapeutic considerations. *Journal of Clinical Ultrasound*, *51*(7), 1223–1230. https://doi.org/10.1002/jcu.23525

Best, R., Gild, A., Huth, J., & Beckmann, J. (2020). Patient-related outcome measurements after operative and conservative management of traumatic proximal adductor longus avulsion injuries. *International Orthopaedics*, *44*(5), 965–971. https://doi.org/10.1007/s00264-020-04510-3

Charnock, B. L., Lewis, C. L., Garrett, W. E., & Queen, R. M. (2009). Adductor longus mechanics during the maximal effort soccer kick. *Sports Biomechanics*, *8*(3), 223–234. https://doi.org/10.1080/14763140903229500

Dijkstra, H. P., Pollock, N., Chakraverty, R., & Ardern, C. L. (2017). Return to play in elite sport: A shared decision-making process. *British Journal of Sports Medicine*, *51*(5), 419–420. https://doi.org/10.1136/bjsports-2016-096209

Dimitrakopoulou, A., Schilders, E. M. J., Talbot, J. C., & Bismil, Q. (2008). Acute avulsion of the fibrocartilage origin of the adductor longus in professional soccer players: A report of two cases. *Clinical Journal of Sport Medicine*, *18*(2), 167–169. https://doi.org/10.1097/JSM.0b013e318164f40b

Harper, D. J., McBurnie, A. J., Santos, T. D., Eriksrud, O., Evans, M., Cohen, D. D., Rhodes, D., Carling, C., & Kiely, J. (2022). Biomechanical and neuromuscular performance requirements of horizontal deceleration: A review with implications for random intermittent multi-directional sports. *Sports Medicine*, *52*(10), 2321–2354. https://doi.org/10.1007/s40279-022-01693-0

Hölmich, P., Uhrskou, P., Ulnits, L., Kanstrup, I.-L., Nielsen, M. B., Bjerg, A. M., & Krogsgaard, K. (1999). Effectiveness of active physical training as treatment for long-standing adductor-related groin pain in athletes: Randomised trial. *Lancet*, *353*(9151), 439–443. https://doi.org/10.1016/S0140-6736(98)03340-6

King, E., Franklyn-Miller, A., Richter, C., O'Reilly, E., Doolan, M., Moran, K., Strike, S., & Falvey, É. (2018). Clinical and biomechanical outcomes of rehabilitation targeting intersegmental control in athletic groin pain: Prospective cohort of 205 patients. *British Journal of Sports Medicine*, *52*(16), 1054–1062. https://doi.org/10.1136/bjsports-2016-097089

Lauersen, J. B., Bertelsen, D. M., & Andersen, L. B. (2014). The effectiveness of exercise interventions to prevent sports injuries: A systematic review and meta-analysis of randomised controlled trials. *British Journal of Sports Medicine*, *48*(11), 871–877. https://doi.org/10.1136/bjsports-2013-092538

Lempainen, L., Hetsroni, I., Kosola, J. A., Sinikumpu, J. J., Mazzoni, S., & Orava, S. (2021). Proximal adductor longus tendon repair with a concomitant distal fascial release for complete hip adductor tendon tears: Surgical technique and outcomes in 40 male athletes. *Orthopaedic Journal of Sports Medicine*, *9*(10), 232596712110420. https://doi.org/10.1177/23259671211042024

Mitchell, A., & Gimpel, M. (2024). A return to performance pathway for professional soccer: A criteria-based approach to return injured professional players back to performance. *JOPST-Open*, (May), 1–42. https://doi.org/10.2519/josptopen.2024.1240

Mitchell, A., Holding, C., & Greig, M. (2021). The influence of injury history on countermovement jump performance and movement strategy in professional soccer players: Implications for profiling and rehabilitation foci. *Journal of Sport Rehabilitation*, *30*(5), 768–773. https://doi.org/10.1123/jsr.2020-0243

Mitchell, A., Holding, C., & Greig, M. (2022). Factors influencing optimum countermovement jump performance and movement strategy in championship professional football players: Implications for player profiling. *Research in Sports Medicine*, *30*(1), 30–40. https://doi.org/10.1080/15438627.2020.1860049

Mitchell, A., Waite, O., Holding, C., & Greig, M. (2023). The development of a return to performance pathway involving a professional soccer player returning from a multi-structural knee injury: A case report. *International Journal of Sports Physical Therapy*, *18*(2). https://doi.org/10.26603/001c.73317

Mosler, A. B., Crossley, K. M., Thorborg, K., Whiteley, R. J., Weir, A., Serner, A., & Hölmich, P. (2017). Hip strength and range of motion: Normal values from a professional football league. *Journal of Science & Medicine in Sport*, *20*(4), 339–343. https://doi.org/10.1016/j.jsams.2016.05.010

Mosler, A. B., Weir, A., Eirale, C., Farooq, A., Thorborg, K., Whiteley, R. J., Hölmich, P., & Crossley, K. M. (2018). Epidemiology of time loss groin injuries in a men's professional football league: A 2-year prospective study of 17 clubs and 606 players. *British Journal of Sports Medicine*, *52*(5), 292–297. https://doi.org/10.1136/bjsports-2016-097277

Serner, A., Jakobsen, M. D., Andersen, L. L., Hölmich, P., Sundstrup, E., & Thorborg, K. (2014). EMG evaluation of hip adduction exercises for soccer players: Implications for exercise selection in prevention and treatment of groin injuries. *British Journal of Sports Medicine*, *48*(14), 1108–1114. https://doi.org/10.1136/bjsports-2012-091746

Serner, A., Weir, A., Tol, J. L., Thorborg, K., Lanzinger, S., Otten, R., & Hölmich, P. (2020). Return to sport after criteria-based rehabilitation of acute adductor injuries in male athletes: A prospective cohort study. *Orthopaedic Journal of Sports Medicine*, *8*(1), 232596711989724. https://doi.org/10.1177/2325967119897247

Serner, A., Weir, A., Tol, J. L., Thorborg, K., Roemer, F., Guermazi, A., Yamashiro, E., & Hölmich, P. (2018). Characteristics of acute groin injuries in the adductor muscles: A detailed MRI study in athletes. *Scandinavian Journal of Medicine & Science in Sports*, *28*(2), 667–676. https://doi.org/10.1111/sms.12936

Thorborg, K. (2023). Current clinical concepts: Exercise and load management of adductor strains, adductor ruptures, and long-standing adductor-related groin pain. *Journal of Athletic Training*, *58*(7–8), 589–601. https://doi.org/10.4085/1062-6050-0496.21

Ueblacker, P., English, B., & Mueller-Wohlfahrt, H.-W. (2016). Nonoperative treatment and return to play after complete proximal adductor avulsion in high-performance athletes. *Knee Surgery, Sports Traumatology, Arthroscopy*, *24*(12), 3927–3933. https://doi.org/10.1007/s00167-015-3669-6